

Syeda Arisha Hassan

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Profile Summary

Software Engineering student at COMSATS Islamabad specializing in AI/ML, LLM fine-tuning, and full-stack development. Fine-tuned and deployed Llama 3 and TinyLlama models on Hugging Face, built a DDoS detection system with 99.57% accuracy, and shipped 10+ projects across ML, mobile, embedded systems, and web. Seeking an AI/ML internship to apply hands-on experience in real-world systems.

Education

COMSATS University Islamabad, Sahiwal Campus – Okara, Pakistan

Sep 2027

Bachelor of Software Engineering

CGPA: 3.51 | Expected Graduation: Sep 2027

Certifications

Python for Data Science, AI & Development by IBM

Grade: 87.75%

Introduction to Python by Data Camp

Completed: Sep 18, 2025

AI & Responsible AI by Microsoft Learn

Fundamentals of Generative AI, Azure OpenAI Service, Responsible AI Principles, Scaling AI

DLSE Program Scholar

Selected for Deep Learning Specialization track to master deep learning and AI concepts

Skills & Technologies

Languages: Python, C++, Java, Dart, Data Structures & Algorithms

Web & Backend: HTML5, CSS3, JavaScript, React.js, Vite, Express.js, Flask, Socket.io

AI / ML: Scikit-learn, XGBoost, PyTorch, GPT-2, TinyLlama, Gemini API, Groq API, Stable Diffusion XL, QLoRA, PEFT, Unsloth, Ollama, FastAPI, Flask, Generative AI

Data & DB: Pandas, NumPy, Plotly, PostgreSQL, MongoDB, Redis, Firebase, Firestore, MySQL

Mobile: Flutter, Android Studio, Provider, Dart

Embedded: Arduino Mega, Arduino Nano, Python Whisper, Embedded C++

Tools: Git, GitHub, Jira, VS Code, Streamlit, Three.js, OpenCV, pdfplumber

Featured Projects

DDoS Attack Detection System

- ML-based network intrusion detection system achieving 99.57% accuracy across 9 traffic categories on the CIC-DDoS2019 dataset.
- Trained XGBoost Classifier on 125,170 network flows with 77 features.
- Built interactive Streamlit dashboard with batch CSV upload, per-class metrics, and confidence scoring.
- Built Python data pipeline with PyTorch for preprocessing and feature engineering on CIC-DDoS2019 dataset, enabling scalable model training on 125K+ network flows.

Tech: Python, XGBoost, Random Forest, Scikit-learn, PyTorch, Streamlit, Pandas

Cognita — AI Teaching Assistant

- Offline-first mobile AI teaching assistant integrating Google Gemini API and a fine-tuned TinyLlama model.
- Fine-tuned TinyLlama 1.1B using QLoRA on 5,030 custom CS teaching conversations; training loss reduced from 2.34 to 0.55.
- Applied 4-bit quantization via BitsAndBytesConfig; deployed model and live demo on Hugging Face Spaces.
- Integrated Google Gemini API as online fallback with offline TinyLlama inference for low-connectivity users.

Tech: Flutter, TinyLlama, QLoRA, Google Gemini API, Hugging Face, BitsAndBytes, Firebase

Medical Language Model Fine-tuning

- Fine-tuned Llama 3 8B on a medical Q&A dataset using QLoRA (4-bit quantization + LoRA adapters) via Unsloth on Google Colab.

- Trained on 33,955 medical examples with only 0.52% of parameters updated, reducing memory usage while maintaining model quality.
- Deployed fine-tuned adapter on Hugging Face Hub for public use.

Tech: *Python, Unsloth, QLoRA, Llama 3, Hugging Face, Google Colab, PEFT, TRL*

Clirix — Your Thinking Layer

- Fully local AI assistant running 100% offline via Ollama with zero data leaving the machine.
- Streams responses word-by-word with full conversation history and multi-model switching (Mistral, Phi-3, Gemma).
- Custom RAG engine with 700-word chunking and 80-word overlap for PDF document intelligence.
- One-click voice input via Google STT with auto-fill and 5 dynamic tone/persona modes (Formal, Casual, ELI5, Bullets, and Technical).

Tech: *Python, Streamlit, Ollama, SpeechRecognition, pdfplumber*

CollabFlow — Real-Time Collaborative Editor

- Full-stack collaborative document editor with glassmorphism UI, 4 dynamic themes, and role-based access controls (Owner, Editor, Viewer).
- Built with React 19, Vite, Tiptap editor, JWT authentication, and bcrypt password hashing.
- Server-side DOMPurify via JSDOM for Stored XSS prevention; 1-second debounced auto-save engine.
- Integrated Google Gemini 2.5 Flash as AI Proofreader with Gen Z tone rewriting and grammar correction.
- Built-in client-side PDF and MS Word exporters with drag-and-drop table support.

Tech: *React 19, Vite, Tiptap, Node.js, JWT, bcrypt, Google Gemini API, MongoDB*

AI Network Detection & Diagnosis System

- Real-time, full-stack dashboard (Python/Flask, JS) utilizing AI pattern detection to identify and visualize network security threats like DDoS and brute-force attacks.
- MongoDB-backed secure authentication with modular Flask architecture and logging/alert system.

Tech: *Python, Flask, MongoDB, HTML, CSS, JavaScript*

Vibe-Sync — AI-Powered Particle Visualization

- Created generative AI system that transforms text descriptions into animated particle visualizations with depth-aware rendering.
- Implemented custom Three.js shaders and latent diffusion models for real-time 3D graphics and glow effects.
- Integrated vision transformers and attention mechanisms for advanced depth estimation and spatial mapping.

Tech: *Three.js, Stable Diffusion XL, ZoeDepth, Custom GLSL Shaders, WebGL, Transformers*

NeuroTick — Learning App

- Innovative educational application combining 3D models with interactive quiz-based learning for STEM subjects including Biology, Chemistry, Physics, Mathematics, and Astronomy.
- Provides immersive learning through multi-level quizzes (Easy/Medium/Hard) and comprehensive progress tracking analytics.
- Built with Flutter using Provider state management, Firebase backend for authentication and data storage, and custom AR for real-time 3D visualizations.

Tech: *Flutter, Dart, Firebase, Firestore, Provider, 3D GLB, Android Studio*

Study Planner — Academic Management Tool

- Built study planner with HTML, CSS, and JavaScript featuring integrated test scheduling system for academic course management.
- Led cross-functional team of 3 members implementing agile methodology, completing all epics and meeting all project deadlines.
- Applied technical communication and task management using Jira and GitHub practices to coordinate sprint planning, code reviews, and documentation.

Tech: *HTML, CSS, JavaScript, Jira, GitHub*

Languages

- English (Fluent)
- Urdu (Native)